Customer Support Ticketer

An application that manages customer support tickets, automatically routes them to agents, and connects similar issues using data structures to improve resolution efficiency.
Functionalities implemented are: Creation of ticket, Assignment of agents, Linking related tickets, Priority escalation.

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Data Structures Used

Linear Data Structure: Queue / Array
1. Used to store and manage incoming tickets in sequential order.
2. Ensures First-In-First-Out processing of unresolved issues.

Non-Linear Data Structure: Graph (Adjacency Matrix)
1. Represents relationships or dependencies between related tickets.
2. Enables linking and tracing connected issues efficiently.

Reason for Choice:
1. Queues ensure First In First Out(FIFO) ticket processing and allows O(1) insertion and deletion at opposite ends using circular array implementation, ensuring efficient memory usage and fair predictable order for support requests.

2. Graphs allow quick traversal and grouping of connected tickets, enabling faster resolution by mapping. They are advantageous over simple lists or trees by forming multiple links among support tickets.

Abstract Data Type (Linear DS) – Queue

Data: Structures for Ticket (Contains customer support ticket details) , Agent(Agent details) , Queue

I. Queue Operations:
   1. initQueue(): Initialize empty queue
   2. enqueue(ticket): Insert new ticket at the rear end
   3. dequeue(): Remove ticket from the front
   4. isQueueFull() / isQueueEmpty(): Check status of queue and return if its full or empty

II. Ticket Operations:
   1. createTicket(): Creates a ticket
   2. assignAgent:() Assign a respective agent to the ticket ID
   3. resolveTicket(): Click to resolve a ticket and update its status
   4. escalatePriority(): Setting a priority for the issue and incrementing its priority
   5. displayTicket(): Displays a certain ticket
   6. displayAllTickets(): Print the current ticket list

Abstract Data Type (Non Linear DS) – Graph

Data:
Structure for Graph that contains adjMatrix[MAX_TICKETS][MAX_TICKETS], and integer numTickets

Operations:
1. initGraph(): Initialize graph with n nodes
2. addLink(): Connect two related tickets
3. showRelatedTickets(): List all tickets linked to the given ticket

Added Functionalities

5. Search by Keyword
To enable users to search tickets based on words or phrases present in either the customer name, or the issue description.

6. SLA Compliance Report
To track resolution performance based on priority based time limits (Service Level Agreements).
The time taken (in hours) is compared with the SLA limit:
High Priority → 4 hrs
Medium → 24 hrs
Low → 72 hrs

Input / Output Screenshots

```
D:\Akshya\1. Sem 2\DSA\MiniProject>gcc main.c ticket.c graph.c -o ticketer

D:\Akshya\1. Sem 2\DSA\MiniProject>ticketer.exe

========= CUSTOMER SUPPORT TICKETER =========
1. Create Ticket
2. Assign Agent
3. Link Related Tickets
4. Escalate Priority
5. Display Ticket by ID
6. Display All Tickets
7. Show Related Tickets
8. Resolve Ticket
9. Exit
Enter your choice: 1
Enter customer name: Akshya
Enter issue: Login failed
Enter priority (1=High, 2=Medium, 3=Low): 1
Ticket #1 created successfully!
Ticket #1 added to queue successfully.
```

```
========= CUSTOMER SUPPORT TICKETER =========
1. Create Ticket
2. Assign Agent
3. Link Related Tickets
4. Escalate Priority
5. Display Ticket by ID
6. Display All Tickets
7. Show Related Tickets
8. Resolve Ticket
9. Exit
Enter your choice: 2
Enter Ticket ID: 1
Enter Agent ID (101-Rahul, 102-Meera, 103-Arjun): 101
Agent Rahul assigned to Ticket #1 successfully.
```

```
========= CUSTOMER SUPPORT TICKETER =========
1. Create Ticket
2. Assign Agent
3. Link Related Tickets
4. Escalate Priority
5. Display Ticket by ID
6. Display All Tickets
7. Show Related Tickets
8. Resolve Ticket
9. Exit
Enter your choice: 3
Enter Ticket IDs to link (e.g. 1 2): 2 3
Ticket #2 and Ticket #3 linked successfully.
```

```
========= CUSTOMER SUPPORT TICKETER =========
1. Create Ticket
2. Assign Agent
3. Link Related Tickets
4. Escalate Priority
5. Display Ticket by ID
6. Display All Tickets
7. Show Related Tickets
8. Resolve Ticket
9. Exit
Enter your choice: 4
Enter Ticket ID to escalate: 3
Ticket #3 priority escalated successfully.
```

Input / Output Screenshots

```
========= CUSTOMER SUPPORT TICKETER =========
1. Create Ticket
2. Assign Agent
3. Link Related Tickets
4. Escalate Priority
5. Display Ticket by ID
6. Display All Tickets
7. Show Related Tickets
8. Resolve Ticket
9. Exit
Enter your choice: 7
Enter Ticket ID to view related tickets: 2

Tickets related to #2: 3
```

```
========= CUSTOMER SUPPORT TICKETER =========
1. Create Ticket
2. Assign Agent
3. Link Related Tickets
4. Escalate Priority
5. Display Ticket by ID
6. Display All Tickets
7. Show Related Tickets
8. Resolve Ticket
9. Exit
Enter your choice: 8
Enter Ticket ID to resolve: 1
Ticket 1 marked as resolved.
```

```
========= CUSTOMER SUPPORT TICKETER =========
1. Create Ticket
2. Assign Agent
3. Link Related Tickets
4. Escalate Priority
5. Display Ticket by ID
6. Display All Tickets
7. Show Related Tickets
8. Resolve Ticket
9. Exit
Enter your choice: 6

-------------------------------------------------
ID | Customer       | Priority | AgentID | Status
-------------------------------------------------
 1 | Akshya         | 1        | 101     | Resolved
 2 | Anagha         | 2        | 102     | Open
 3 | Advitha        | 2        | 103     | Open
-------------------------------------------------
```

```
========= CUSTOMER SUPPORT TICKETER =========
1. Create Ticket
2. Assign Agent
3. Link Related Tickets
4. Escalate Priority
5. Display Ticket by ID
6. Display All Tickets
7. Show Related Tickets
8. Resolve Ticket
9. Exit
Enter your choice: 5
Enter Ticket ID to view details: 1

--- Ticket Details ---
Ticket ID: 1
Customer Name: Akshya
Issue: Login failed
Priority: 1
Agent ID: 101
Status: Resolved
```

Input / Output Screenshots of Added Functionalities

```
========= CUSTOMER SUPPORT TICKETER =========
1. Create Ticket
2. Assign Agent
3. Link Related Tickets
4. Escalate Priority
5. Display Ticket by ID
6. Display All Tickets
7. Show Related Tickets
8. Resolve Ticket
9. Exit
10. Generate SLA report
11. Search Tickets by Keyword
Enter your choice: 11
Enter keyword to search: payment

=========== SEARCH RESULTS ===========

Ticket ID: 1
Customer: Akshya
Issue: Payment issue
Priority: 2
Resolved: No
======================================
```

```
========= CUSTOMER SUPPORT TICKETER =========
1. Create Ticket
2. Assign Agent
3. Link Related Tickets
4. Escalate Priority
5. Display Ticket by ID
6. Display All Tickets
7. Show Related Tickets
8. Resolve Ticket
9. Exit
10. Generate SLA report
11. Search Tickets by Keyword
Enter your choice: 10

=========== SLA COMPLIANCE REPORT ===========
ID | Priority | Status      | Time Taken (hrs)
----------------------------------------------
 1 | 2        | Met         | 0.52
 2 | 2        | Met         | 0.01
----------------------------------------------
```

Conclusion & Team Contribution

Conclusion:
1. Developed a modular C based Customer Support Ticketing System integrating both Queue and Graph DS.
2. Implemented efficient ticket lifecycle management, including creation, assignment, linking and escalation
3. Designed a Graph based relationship model to manage ticket dependencies for faster issue tracking.
4. Demonstrated data structure driven problem solving and the practical use of algorithms in real world service management systems.

Team Contribution:
Member 1: Sivagami
Implemented Queue-based ticket management including creation, enqueue/dequeue, and priority escalation.
Member 2: Anagha
Developed Graph module for linking and displaying relationships between related tickets.
Member 3: Advitha
Handled system integration, menu driven control flow, and testing of all functionalities.

Added functionalities was done together in the lab.